ARXA - Prediction has value through a feasible action

Private portfolio design study. Not a client deployment.

Problem and decision user

An industrial analytics initiative can optimize a prediction metric without improving the operational decision or economics. The proposed user is operations manager, asset owner and finance sponsor.

Operating and economic approach

A model for managing data-enabled operational transformation and investment choices. The package joins the business question, product scope, operating handoffs, technology/data assumptions and financial consequences. Existing tools and simpler alternatives remain part of the decision.

Checked result

Eight tests and twelve efficacy/capacity scenarios show how intervention assumptions can reverse a favorable cost comparison. The act/defer calculation is a cost comparison, not a real-option price.

Capabilities illustrated

Industrial analytics; action constraints; economic scenarios; staged investment. These are demonstrated documentary/model mechanisms, not a certification of personal independent mastery.

Limits and next use

Distribution shift, sensor quality and intervention effectiveness may invalidate estimated savings. Use the detailed specification and validation memo to scope a real mandate, identify missing data and define the smallest decisive test. Publication and deployment require separate review.